

Long term durability of pultruded polymer composite rebar in concrete environment



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ABSTRACT

Pultruded glass fibre reinforced polymer (GFRP) composite rebar was immersed in alkaline concrete environment for 0, 1, 2, 3, 4, 6, 14 and 24 months at 60 °C to evaluate its durability in concrete structure. Moisture absorption of the rebar was found to be only 0.76%. It was also found that both the glass transition temperature (T_g) and the short beam shear strength were retained by about 91.5%; and the above properties were remained almost unchanged during the ageing period from 1 month to 24 months. Design tensile strength and tensile elastic modulus of the rebar were retained by 100% after 24 months exposure in concrete environment. Degradation of GFRP rebar was not evident in the FT-IR results as supported by scanning electron micrographs and energy dispersive X-ray analysis. Moisture absorption was found to be a critical factor that controlled thermal and mechanical properties of GFRP rebar.

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1. Introduction

The use of GFRP composite bar as reinforcement for concrete is still a relatively new technology, and currently there is no standard for the durability properties of GFRP rebar. Handful of information regarding the durability of GFRP rebar in concrete, a highly alkaline environment, is one of the issues limiting the acceptance of this material in civil engineering applications.

Typically, GFRP rebar consists of reinforcing glass fibre which is the hardest, strongest and stiffest component, embedded in a continuous polymer matrix (e.g. polyester, vinyl ester and epoxy resins). The main functions of the matrix are to transmit externally applied loads to the reinforcement and to protect the latter from external mechanical and environmental damages [1]. GFRP rebar do not corrode by electro-chemical reaction as in rusting of steel; however, their durability primarily depends on the constituent materials (e.g. glass and resin types), manufacturing process and structural integrity of the rebar (e.g. no internal crack). Also, strength and stiffness of GFRP rebar may increase, decrease, or remain the same, depending on the particular material and environmental conditions [2].

The polymer matrix of GFRP rebar provides a chemical barrier for the glass fibres from the corrosive effects of moisture and alkalis in concrete. In moist environment, water molecules react with the ester group ($-O-CO-$) of polymer matrix (e.g. polyester resin) which cause irreversible chemical change by hydrolytic reaction

and diminishes toughness and fracture strain. Vinyl ester resin is less vulnerable to hydrolytic reaction because of less ester group present in the polymer molecule compared to polyester resin. Epoxy resin is also resistant to chemical attack; however, price of this resin is reasonably higher than the vinyl ester resin. So vinyl ester resin is being commonly used for manufacturing of GFRP rebar. Both vinyl ester resin and epoxy resin are permitted for fibre reinforced rebar products for permanent applications in concrete [3]. In the case of alkali attack to glass fibre, the chemical reaction involves a breakdown of the silica ($-Si-O-Si-$) network by hydroxide ions (OH^-) of concrete pore solution and eventually glass fibres lose weight and strength when they are in contact with strong alkalis. The rate of deterioration of the fibres depends on the alkalinity of the matrix (i.e. the availability of OH^- ions). Boron free E-CR glass ("Electrical" grade "Corrosion Resistant" glass) is more resistant to strong alkalis than E-glass as reported in [4]. Thus a GFRP rebar made up of chemical resistance resin (e.g. vinyl ester resin or epoxy resin) along with corrosion resistant glass is expected to be highly durable in concrete environment subject to the facts that the resin of the rebar is well cured and the rebar is free from cracks and internal flaws.

Preferably, GFRP rebar that is intended for long term use should be examined in real time and in-service environment. The required service life of concrete reinforcement is usually in the order of 50–100 years. However, the time involved is not viable and therefore, accelerated ageing (or weathering) techniques are required. The most common approach to accelerate ageing uses elevated temperature tests combined with reasonable in-service environment

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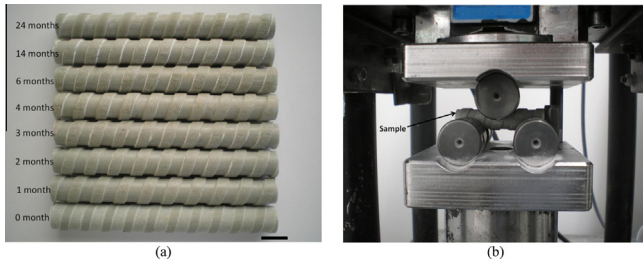


Fig. 1. Photographs of (a) alkaline aged and control rebar samples (scale = 15 mm) and (b) typical short beam shear test setup.

[5]. Short term accelerated ageing has often been extrapolated to make long term mechanical property estimates [5,6]. It has been reported that 42 days immersion of GFRP in alkaline environment (pH $\approx$ 13) at 60 °C would corresponds to 28 years in a real concrete structure [7]. In another report, Deijke [8] predicted that conditioning of GFRP rebar for 18 months in concrete at 60 °C corresponds to 100 years at outdoor conditions where the mean annual temperature is 7 °C.

A number of investigations have been conducted to evaluate the effects of moisture, loading, temperature, polymer matrix and different harsh environments on the properties of GFRP rebar [7,9–11]. However, there have not been many investigations that have explained properties of GFRP rebar over long periods of time, a year or longer. Objective of the entire research project, of which this work is a part, is to systematically investigate long term durability of pultruded GFRP rebar in concrete environment. In particular, this paper presents the effects of long period (24 months) alkaline ageing at elevated temperature on thermal and mechanical properties of GFRP rebar.

2. Materials and methods

2.1. Material

The GFRP rebar was received from Pultron Composites Limited, New Zealand. These threaded rebar (brand name is Mateenbar™) of 14 mm diameter (outer diameter) were made up of unidirectional roving of E-CR glass and epoxy vinyl ester resin as manufactured through pultrusion process. The GFRP rebar samples were well cured and crack free.

2.2. Methods

2.2.1. Alkali resistance test

A temperature controlled stainless steel tank was used to immerse the GFRP bars in alkaline solution for 1, 2, 3, 4, 6, 14 and 24 months at 60 °C. Typical alkaline solution was made up of 118.5 g of Ca(OH)₂, 0.9 g of NaOH, and 4.2 g of KOH in 1 L of deionised water [12]. The alkaline solution had a pH value of around 13, a representative pH value of mature concrete pore solution. Photograph of alkaline aged rebar samples is presented in Fig. 1a.

2.2.2. Moisture absorption test

Moisture absorption of the rebar was monitored periodically from 1 h to 17280 h (i.e. 24 months). A digital scale with accuracy of ± 0.0001 g was used to weight the samples. Water uptake of three samples was recorded in each time and the average values are presented. The percentage moisture uptake was measured through gravimetric means using the following equation:

$$M(\%) = \{(W_t - W_o)/W_o\} \times 100 \quad (1)$$

where W_t is the weight of the wet specimen and W_o is the weight of the dry specimen.

Precautions were taken to remove the surface moisture from all the rebar specimens by carefully wiping them off each time before weighing.

2.2.3. Fourier transform infrared (FT-IR) spectroscopy

A Nicolet 6000 FT-IR from Thermo Scientific was used to obtain spectra of controlled and alkaline aged samples. The potassium bromide (KBr) disc sample preparation method was followed to take the infrared spectra. The samples were ground and mixed with KBr at a ratio of 1:99. Then the mixer was pressed under vacuum to form pellets. FT-IR spectra were recorded in a range of 4000–650 cm⁻¹ at a resolution of 4 cm⁻¹.

2.2.4. Differential scanning calorimetry (DSC)

T_g measurement was carried out using a TA Instruments Modulated DSC 2920. About 15–20 mg sample was scanned at a heating rate of 5 °C/min from –40 to 150 °C with a modulation period of 60 s and a modulation peak of ± 0.5 °C. T_g was evaluated by analysing the heat flow signal using TA Instruments' Universal Analysis software package and it was identified as the temperature at half height of the step change. Three samples of each type were tested and the average value is reported.

2.2.5. Short beam shear test

In this work, short beam shear test was performed using test configuration as described in ASTM: D4475. Typical test set-up is shown in Fig. 1b. Short beam shear strength (SBSS) was calculated using the following equation:

$$SBSS = 0.849 \times (P/d)^2 \quad (2)$$

where P is the breaking load and d is the diameter of specimen. Five samples of each type were tested and the average value is reported.

2.2.6. Tensile test

Specimen preparation and tensile test were conducted according to the guidelines as specified in ASTM: D7205/D7205M and ACI 440.3R-04 [12]. Tensile specimen for testing was prepared by anchoring two ends of the rebar in steel plugs filled with epoxy resin. The free length between the steel plugs was about 40 times of root diameter of the rebar. The tensile test was performed on 4–6 replicates of each batch using a universal testing machine (WDW-200E) fitted with 200 kN load cell. Load ramp rate was set at 667 N/s, and an extensometer of 50 mm gauge length was mounted with clips at the centre of test specimen. Load versus extension data were acquired by computer, which was also facilitated machine control. The ultimate tensile strength, elastic modulus and tensile strain were calculated using Eqs. (3), (4) and (5), respectively:

$$f_u = F_u/A \quad (3)$$

$$E_L = (F_1 - F_2)/\{(\varepsilon_1 - \varepsilon_2) \times A\} \quad (4)$$

$$\varepsilon_u = F_u/(E_L \times A) \quad (5)$$

where f_u is the tensile strength, MPa; F_u the tensile capacity, N; A the cross-sectional area of root diameter, mm²; E_L the longitudinal modulus of elasticity, MPa; F_1 and ε_1 the load and corresponding strain, respectively, at approximately 50% of the ultimate tensile capacity, N and dimensionless, respectively; F_2 and ε_2 are the load and corresponding strain, respectively, at approximately 20% of the ultimate tensile capacity, N and dimensionless, respectively; ε_u = ultimate strain, dimensionless.

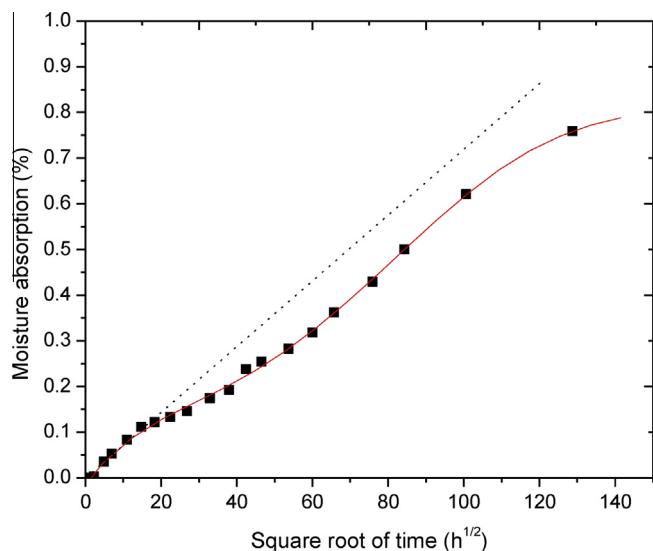


Fig. 2. Moisture absorption of GFRP rebar as a function of square root of alkaline ageing time.

2.2.7. Scanning electron microscope (SEM) and Energy dispersive X-ray (EDX)

The effect of alkaline ageing on the morphology of fibre/matrix interface of GFRP rebar samples was investigated using a scanning electron microscope (Model: MV2300) from CamScan Electron Optics. Elemental analysis of glass fibre of GFRP rebar was conducted by EDX. Samples were sputter coated with gold to make them conductive prior to SEM and EDX analysis.

3. Results and discussion

3.1. Moisture absorption

Polymer composites tend to absorb moisture when exposed to high humidity. The surface layers of composites immediately start absorbing moisture in contact with the aqueous environment. However, moisture diffusion into the bulk material is usually a slow process. It may take weeks to months before a substantial amount of moisture has been absorbed by the composite, and considerably longer periods (i.e. >1 year) before the material is saturated [13]. Fig. 2 shows the average moisture absorption of GFRP rebar in alkaline environment (pH $\approx$ 13) at 60 °C with respect to square root of immersion time ($\sqrt{t}$). Several features were noted in the figure. First, the moisture uptake increased steadily with respect to $\sqrt{t}$ as expected by Fick's law. However, no equilibrium uptake level was reached up to 24 months of ageing period. After the fast diffusion, the moisture uptake continued to increase gradually with respect to $\sqrt{t}$, which cannot be explained by Fickian diffusion. The long term moisture absorption trend as shown in Fig. 2 was considered as two stages non-Fickian [14]. This non-Fickian (or non-equilibrium) moisture absorption behaviour could be due to (i) increase in free volume in polymer as a result of gradual relaxation of the resin (see Section 3.3), (ii) binding of water molecules to the molecular structure of the resin and (iii) capillary transport of water molecules to fibre/matrix interfaces.

Secondly, the moisture uptake of the rebar at 60 °C was found to be only 0.76% by weight after 24 months of immersion in alkaline solution, a finding lower than other alkaline moisture uptake results as reported in [4,15]. The above low moisture absorption indicated that there were no internal flaw or path for diffusion and accumulation of excessive moisture. This was also an indication of good interfacial bonding between glass fibre and resin of the

Table 1

FT-IR transmittance peaks of the alkaline aged and control GFRP rebars.

Possible assignment	Transmittance peak (cm ⁻¹)
OH stretching vibrations	3400
C–H stretch of the benzene ring	3026
CH, CH ₂ and CH ₃ stretch	2960/2927/2871
C=O stretch	1722
Aromatic ring stretch	1605/1581/1506
Aromatic ring stretch	1448
CH ₂ and CH ₃ scissor	1380
C–O stretch	1295
C–CO–C stretch	1160
C=C stretch	920
Aromatic ring stretch or CH ₂ rocking	755

GFRP rebar, and consistency in glass fibre distribution across the rebar.

3.2. Fourier transform infrared spectroscopy

In general, vinyl ester is a polymerisation product of methacrylic acid and diglycidyl ether of bisphenol A (DGEBA). For processing purpose, vinyl ester is dissolved in a monomer, or reactive diluent, usually styrene, the result is a low viscosity liquid referred to as resin. Obviously the styrene is not only used as solvent, but actively takes part in the cure reaction to form a cross-linked vinyl ester polymer (see Fig. 3). Vinyl esters contain only one ester group at each end of the polymer molecule. Also methyl group occupies a lot of space and sterically hinders any molecule approaching the ester group by impeding their access. These two aspects of the design of the vinyl ester molecule combine to make them chemically resistant.

Fig. 4 shows FT-IR spectra of control and 24 months alkaline aged GFRP rebar samples. In general, the spectrum of alkaline aged sample was very similar to that of the control sample (see Fig. 4a). A summary of the characteristic peaks is presented in Table 1. Ester group of vinyl ester is susceptible to hydrolysis during ageing and this process is accelerated and catalysed by the presence of acids or bases. The change in the relative peak intensity of hydroxyl (OH) group before and after ageing provides insight into the hydrolysis reaction of the sample. The extent of hydrolysis can be measured by determining the ratio of the maximum peak intensity of OH group at 3400 cm⁻¹ and the peak intensity of carbon-hydrogen (C–H) of aromatic group due to stretching at 3026 cm⁻¹ (see Fig. 4b). It was assumed that aromatic group was not degraded, so the C–H content of aromatic group was remained constant [16]. It was found that relative peak intensity of OH group of alkaline aged sample was changed only by 0.3% when compared with that of control sample. This finding suggested that hydrolysis occurred but was superficial; the majority of the specimen was not degraded. Since the degree of hydrolysis of 24 months alkaline aged sample was insignificant, so it is likely that other alkaline aged samples (1 to 14 months) was not hydrolysed significantly. In a field study report, Mufti et al. [16] also found an insignificant change in FT-IR spectra of GFRP bars embedded in concrete for 6–8 years.

3.3. Glass transition temperature

Fig. 5 shows typical DSC heating thermograms of control (0 month) and alkaline aged samples. In general, the curves showed a declining step in heat flow due to glass transition in the range of 93–106 °C. It also appeared that glass transition step of 0, 1, 2, 3 and 4 months alkaline aged samples was not quite intense (see Fig. 5a) when compared with that of 6, 14 and 24 months alkaline aged samples (see Fig. 5b). The above intense step of long period (6,

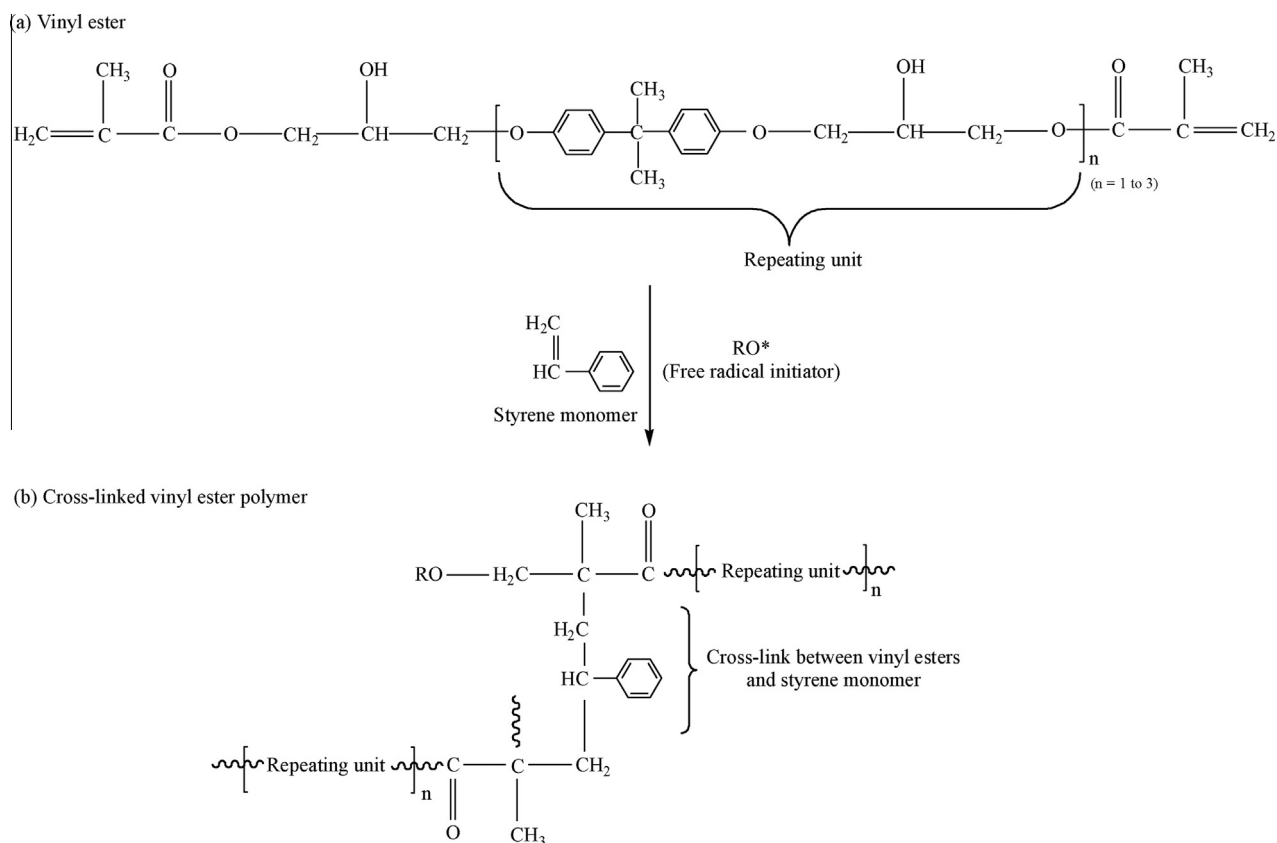


Fig. 3. Schematic chemical structures of (a) vinyl ester and (b) cross-linked vinyl ester polymer.

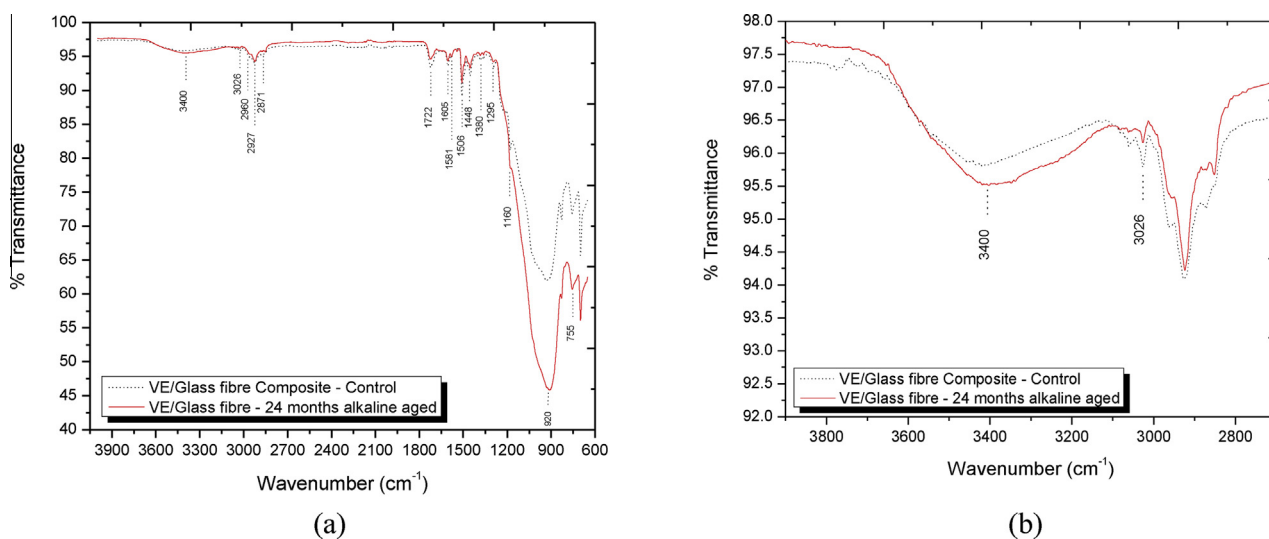


Fig. 4. FT-IR spectra of control and alkaline aged GFRP rebar samples.

14 and 24 months) alkaline aged samples was appeared as an endothermic or melting peak at T_g ; but the transformation was not involved melting process whatsoever. This peak was attributed to the thermal stress relaxation phenomenon of polymer chains of the GFRP rebar during long term ageing. It was believed that thermal stress was induced in polymer chains during long period hygrothermal alkaline ageing. DSC traces also indicated that there was no residual curing of the samples during heat scanning.

Fig. 6 shows the average T_g of GFRP rebar as a function of alkaline ageing time. As can be seen, the average T_g of control sample is

105 °C and it decreased to a minimum of about 96 °C in the first month of immersion. With a further increase of immersion (or ageing) time up to 24 months, T_g of the GFRP rebar did not change remarkably. The overall drop of T_g by about 9 °C was attributed to the absorbed moisture as shown in Fig 2. It is well known that absorption of moisture causes plasticization of the polymer to occur with concurrent swelling and lowering of the polymer's glass transition temperature [17]. The absorbed water molecules increased the distance between the polymer segments resulting in an increase in chain mobility (or free volume) and thus a decrease in the T_g .

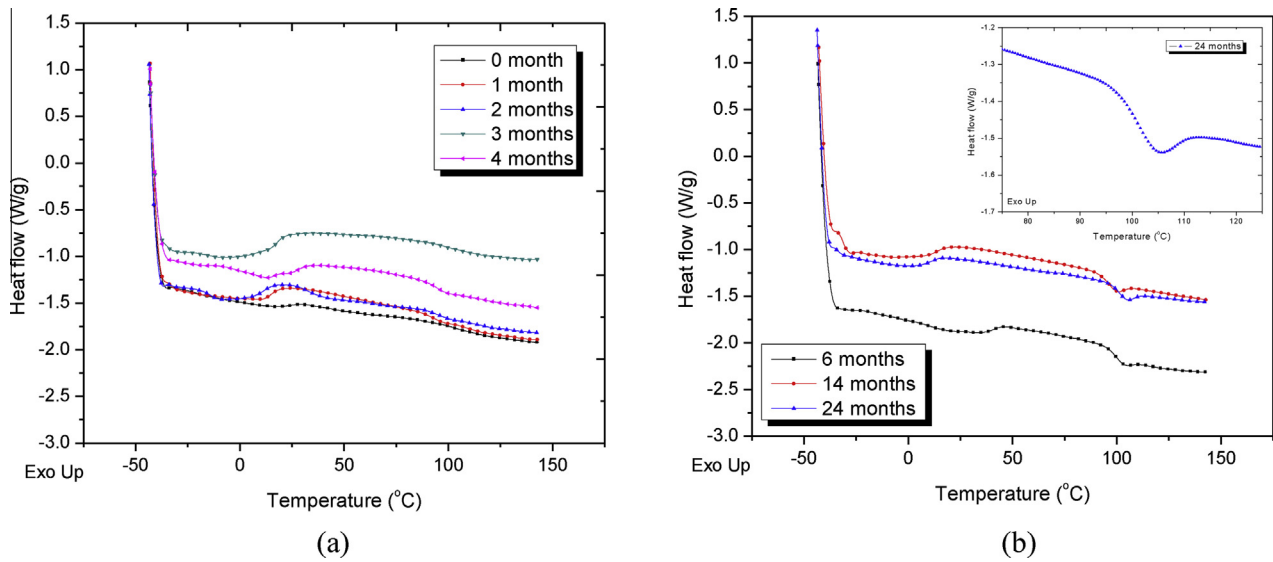


Fig. 5. DSC thermograms of control and alkaline aged GFRP rebar samples.

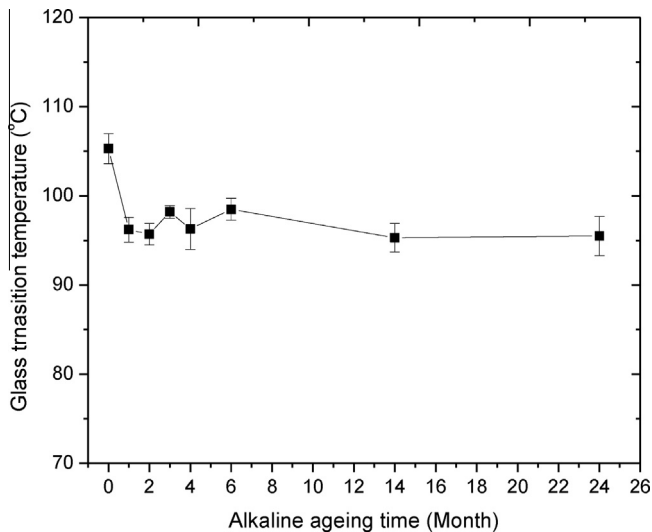


Fig. 6. Glass transition temperature of GFRP rebar as a function of alkaline ageing time.

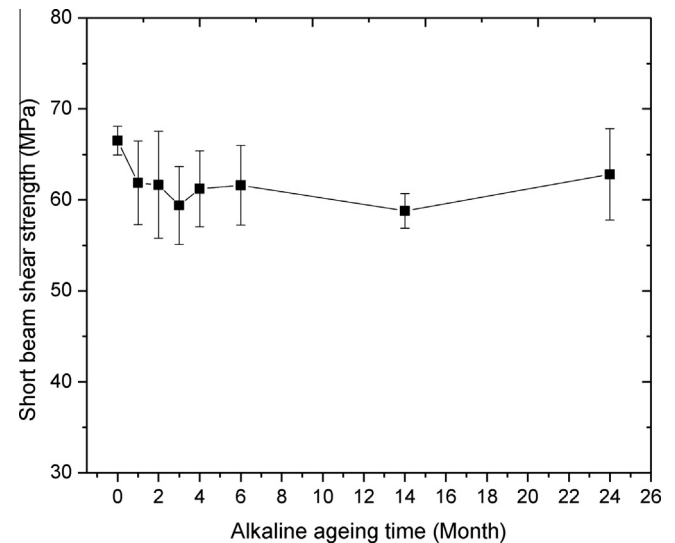


Fig. 7. Short beam shear strength of GFRP rebar as a function of alkaline ageing time.

3.4. Short beam shear strength

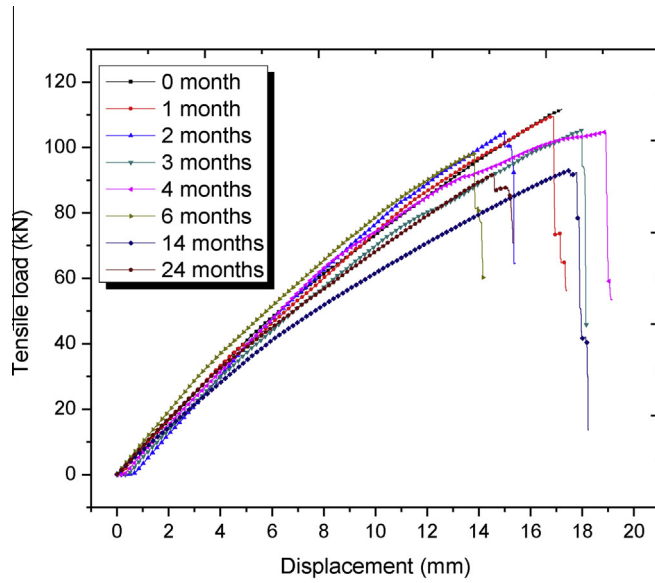
The effect of alkaline ageing on SBSS of GFRP composites is not well documented yet. SBSS of GFRP composite has been focused only in little researches. For instance, SBSS of two types of GFRP rebars fabricated with E-glass and vinyl ester resin was reported by Chen et al. [18]. The samples were immersed in alkaline conditions for 45 days at 60 °C. In this work, SBSS of the samples was decreased by 8–20% due to alkaline ageing. In another work, Micelli and Nanni [7] reported that SBSS of E-glass/polyester rebar decreased by more than 90% due to immersion in alkaline solution after 42 days at 60 °C. This significant loss of SBSS was attributed to the large amount of absorbed moisture by E-glass/polyester rebar.

The average SBSS of the control and the alkaline aged samples with respect to time is presented in Fig. 7. Like T_g , SBSS of the control samples was dropped from 66.5 MPa to 61 MPa due to alkaline ageing; however, SBSS was quite steady with respect to time for all the alkaline aged samples. The overall drop in SBSS of alkaline aged

samples was recorded to be around 8% which was consistent with some extent with the findings as reported in [18]. This reduction in SBSS was again attributed to absorbed moisture as discussed earlier (see Section 3.1). The failure of all the samples was clearly a horizontal shear failure began at mid-span and propagated to the left or right with no preference along the neutral axis as observed during testing.

3.5. Tensile properties

Typical tensile load versus displacement curves of the control and the alkaline aged rebar samples are presented in Fig. 8a. As can be observed, the curves initially showed linear deformation, followed by a slight non-linear deformation prior to attainment of the maximum load. Once the load reached maximum, the recorded load dropped suddenly. Overall, the tensile failure of GFRP rebar was neither completely brittle nor completely ductile, but rather mixed. It also appeared that load–displacement curves of



(a)



(b)

Fig. 8. (a) Typical tensile load versus displacement curves of GFRP rebar and (b) photograph of typical tensile failure of GFRP rebar by splintering.

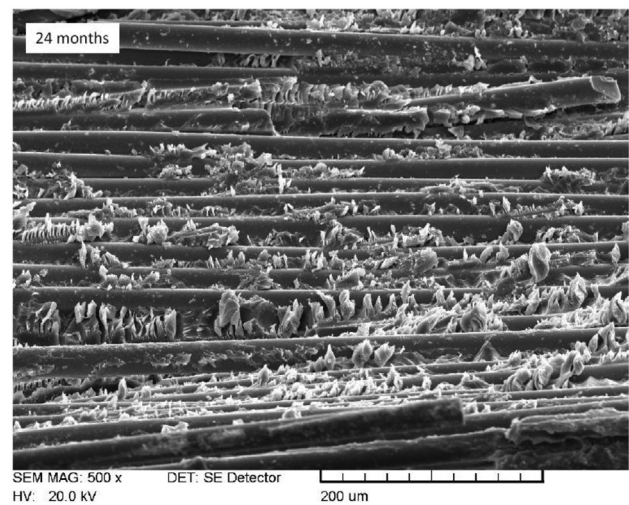
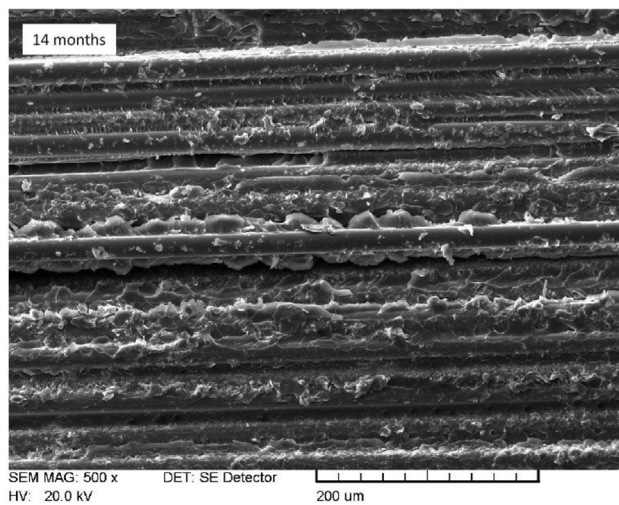
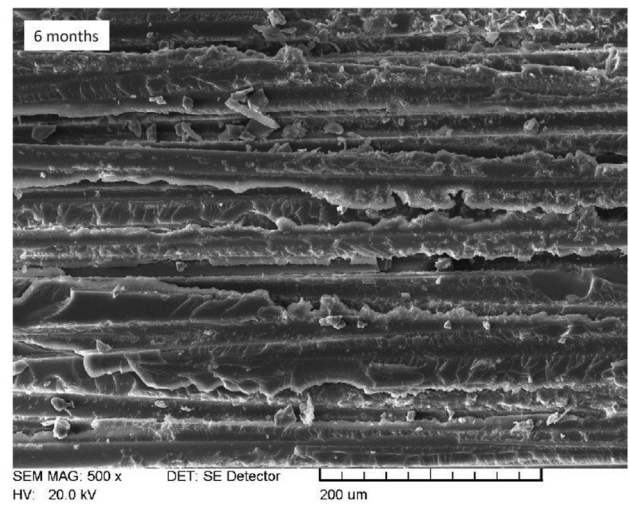
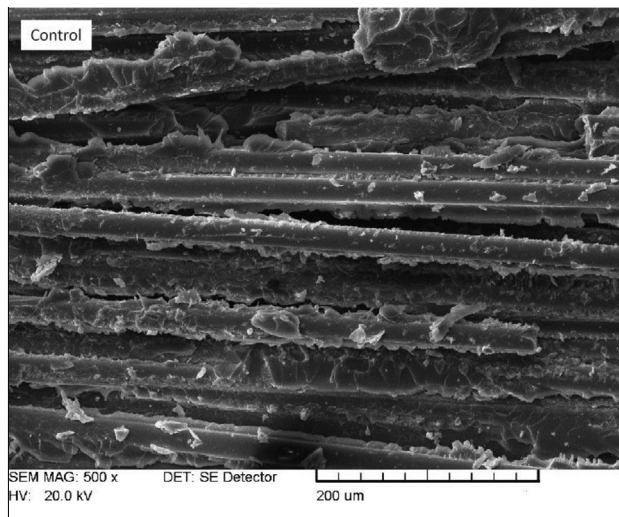


Fig. 9. SEM micrographs of tensile fracture surface of control and alkaline aged GFRP rebar samples.

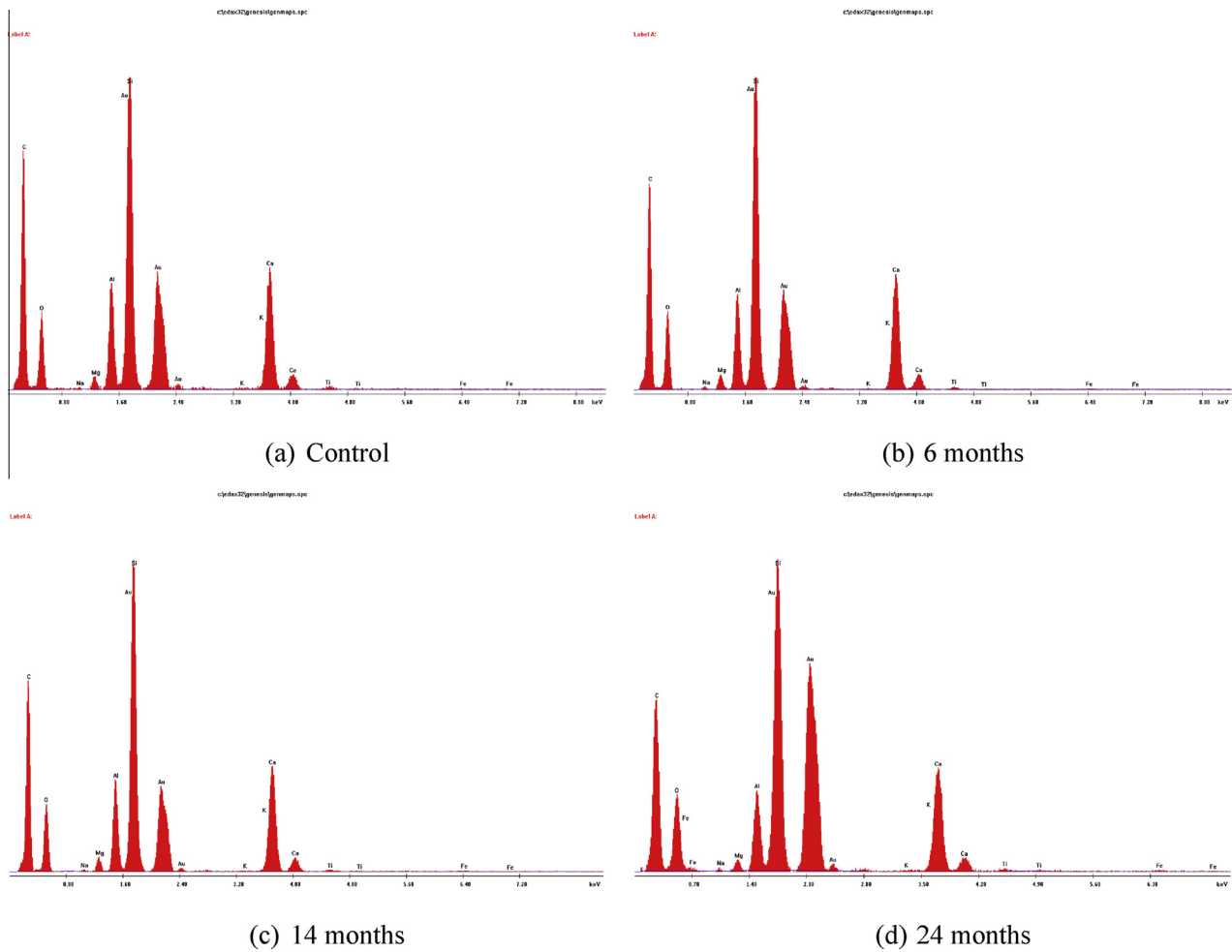


Fig. 10. Typical EDX spectra of control and alkaline aged GFRP rebar samples.

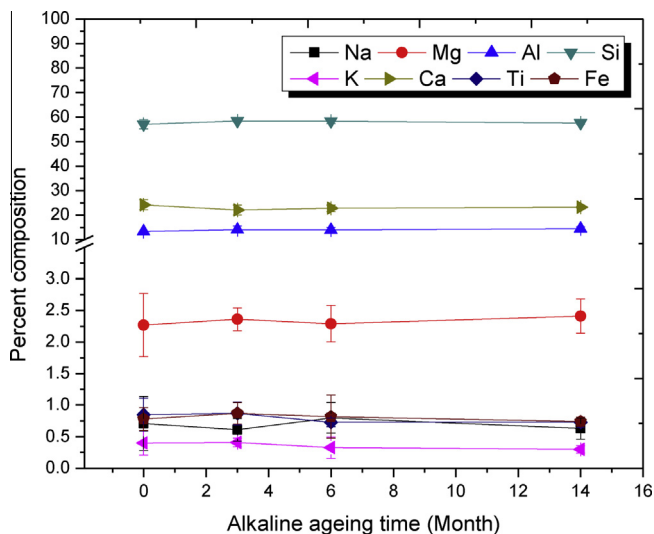


Fig. 11. Percent composition of elements of glass fibre of GFRP rebar as a function of alkaline ageing time.

the alkaline aged rebars did not change significantly compared with that of the control rebar.

All the rebar samples were failed in tensile test by splintering along the specimen free length (see Fig. 8b) as a result of combined

transverse compression and longitudinal extension in the composites. SEM micrographs (see Fig. 9) of tensile fracture surfaces showed that fibres were well connected with the matrix and the fibre/matrix interface was not affected remarkably due to exposure in an alkaline environment for a long period. In all the cases, there were signs of matrix adhering to the fibres which indicated good bonding at the fibre/matrix interface. Additional investigation regarding alkaline attack on the glass fibre (i.e. corrosion of glass fibre) of tensile tested GFRP rebar was conducted using EDX as shown in Fig. 10. As can be seen, the glass fibre was mainly composed of SiO_2 , Al_2O_3 , CaO , MgO , Na_2O , K_2O , TiO_2 and Fe_2O_3 . If the glass fibre degraded during ageing due to dissolution, leaching and electrochemical reaction of metal ions of glass and OH^- ions of the alkaline solution, then it could be expected that the composition of glass fibre was also changed quantitatively. Fig. 11 shows that fibre glass compositions of alkaline aged rebars were not changed significantly when compared with that of the control rebar sample. The variations in glass fibre compositions for different ageing periods were within the standard deviation limit.

The design of concrete structure reinforced with GFRP rebar shall satisfy strength and serviceability requirements. According to American Concrete Institute guidelines [12], the design tensile strength shall be determined using the following equation:

$$f_u^* = (f_{u,avg} - 3\sigma) \times C_E \quad (6)$$

where f_u^* is the design tensile strength, $f_{u,avg}$ is the average tensile strength, σ is the standard deviation and C_E is the environmental

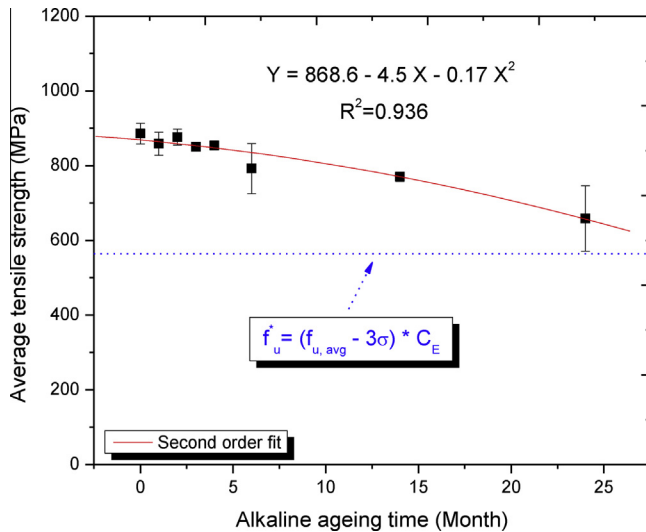


Fig. 12. Tensile strength of GFRP rebar as a function of alkaline ageing time.

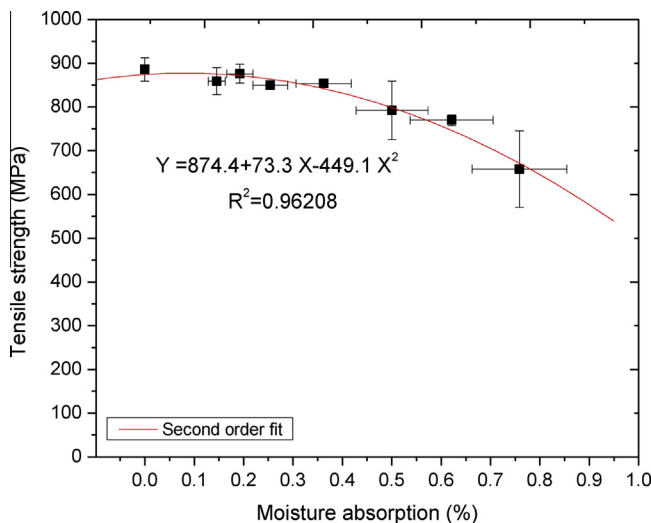


Fig. 13. Tensile strength versus alkaline moisture absorption of GFRP rebar.

reduction factor. C_E is considered as 0.7 for GFRP rebar exposed to earth and weather, and accounts for the expected loss in tensile strength of the GFRP rebar in an alkaline environment over the course of its service life [3]. The average tensile strength of GFRP rebar as a function of alkaline ageing time is presented in Fig. 12. The tensile strength of control rebar was found to be 886 MPa (± 27 MPa); and the design tensile strength of the control rebar was 563 MPa as estimated using Eq. (6).

Again in Fig. 12, tensile strength of GFRP rebar slowly decreased with time as anticipated. The decrease in tensile strength (about 26%) was due to weakening of fibre/matrix interface as a result of absorption of moisture during ageing. Fig. 13 shows a strong relationship ($R^2 = 0.962$) between the tensile strength and the moisture absorption of GFRP rebar, where tensile strength decreased with increased moisture absorption. It is worth to mention here that properties of GFRP composite deteriorate much faster in alkaline solution at high temperature (i.e. accelerated ageing) than in concrete due to easy mobility and diffusion of moisture and ions of the solution through the matrix to the fibre/matrix interface. It is shown in Fig. 12 that tensile strength of GFRP rebar was above the design tensile strength even after 24 months of accelerated

Table 2

Tensile elastic modulus and tensile strain of alkaline aged and control GFRP rebars.

Alkaline ageing time (Month)	Tensile elastic modulus (GPa)	Tensile strain (%)
0	55.03 (± 1.39)	1.62
1	56.48 (± 0.83)	1.51
2	54.76 (± 1.60)	1.60
3	53.38 (± 1.77)	1.58
4	54.35 (± 1.53)	1.56
6	54.13 (± 1.42)	1.48
14	54.74 (± 1.93)	1.40
24	57.24 (± 0.53)	1.15

ageing, which suggested that tested GFRP rebar was durable in alkaline environment.

Dejke [8] investigated durability of alkali resistance glass and E-glass reinforced vinyl ester and polyester rebars. It was found that tensile strength of all types of rebars dropped by about 50% after 1 year exposure in concrete at 60 °C. It was also observed that alkaline solution is more aggressive environment than concrete at 100% relative humidity for GFRP rebars. In another work, Benmokrane et al. [4] reported that, under a 30% stress level and 140 days ageing in high pH pore water solution and in concrete, tensile strength of alkali resistance glass/vinyl ester rebar was not changed noticeably. On the other hand, tensile strength of E-glass/vinyl ester decreased by 16–17%. Masmoudi et al. [19] exposed E-glass/vinyl ester rebars in deionised water and alkaline solution under a stress level of 20–29% for 104 days with temperature varying from 45 to 63 °C. They found that tensile strength was dropped by 15% and 17% due to exposure in deionised water and alkaline solution, respectively. In another research, Davalos et al. [20] found that tensile strength of E-glass/vinyl ester rebars decreased by about 55% after 270 days conditioning in concrete and water at 60 °C. The discrepancies in tensile strength of GFRP rebar as reported in literatures and in present work were mainly due to variation in constituents of GFRP rebar, manufacturing method and test conditions.

The average tensile elastic modulus results of control and alkaline aged GFRP rebars are presented in Table 2. The tensile elastic modulus of GFRP rebar was found to be quite stable up to 24 months of ageing in alkaline environment. It was ranged from 53 to 57 GPa irrespective to ageing time. Statistical analysis (t -test at a confidence level of 95%) did not support a significant difference in tensile modulus of alkaline aged rebars in relation to that of the control rebar. Again in Table 2, tensile strain of the composites showed a general declining trend with respect to alkaline ageing time, which was due to change in tensile strength as described by Eq. (5).

4. Conclusions

In this research, the effect of long term alkaline ageing (i.e. concrete environment) on the thermal and mechanical properties of GFRP rebar has been investigated. The alkaline ageing of GFRP rebar was conducted at elevated temperature (60 °C) which was considered as accelerated ageing test. The moisture absorption of GFRP rebar as studied in this work was found to be very little after 24 months of ageing. In addition, it was revealed that moisture absorption of GFRP rebar followed two stages non-Fickian process. This was due to long term diffusion of water molecules within the rebar at elevated temperature as a result of relaxation and increase in free volume of polymer chain, and accumulation of water molecules at the fibre/matrix interface. Vinyl ester polymer of GFRP rebar was not degraded notably in high alkaline concrete environment as confirmed by FT-IR results. T_g results also supported

that polymer matrix was not degraded notably, whereas about 9 °C reduction in T_g of alkaline aged rebar was due to plasticization of the polymer by water molecules. SBSS was decreased by 8% but appeared reasonably stable in alkaline environment, a trend similar to T_g . The reduction in shear strength was attributed again to the absorbed moisture of the rebar. The tensile strength of the rebar was retained by 74%, and the tensile elastic modulus was not changed significantly after 24 months of ageing. Tensile elastic modulus of the rebar is a function of glass fibre content; on the other hand, tensile strength is associated with glass fibre content and fibre/matrix bond strength. During the alkaline ageing period, glass fibre was not degraded as confirmed by SEM and EDX, and accordingly the tensile elastic modulus was remained almost unchanged; however, tensile strength of GFRP rebar decreased as a result of weakening of load transfer efficiency from the polymer matrix to the fibre due to absorption of moisture at the fibre/matrix interface via capillary diffusion.

It is understandable that polymer composites, in general, absorb moisture to some extent from the environment. So design engineers should give emphasis on moisture absorption property along with other thermal and mechanical properties of GFRP rebar for structural applications. GFRP rebar for strengthening concrete structure should be resistance to moisture by structural integrity (i.e. internal/external defects or other wicking path) and the resin should also be moisture resistant. It is important to note that a significant amount of moisture could be absorbed even by a moisture resistant resin based GFRP rebar if defect and flaw are exist, and may cause material swelling and, particularly at higher temperatures, may soften and weaken the polymer matrix and fibre/matrix interface, which is deleterious to mechanical properties that are often design drivers for structural applications.

The pultruded GFRP rebar studied in this work was found to be durable in alkaline/concrete environment due to very little moisture absorption and so the tensile strength of the rebar was well above the design tensile strength even after 24 months exposure in high temperature alkaline environment. More data on tensile strength of GFRP rebar at different temperatures is required to develop a prediction model which will be reported elsewhere.

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